

Quantitative Biomechanical Benchmarks for Youth Baseball Infield Throwing Mechanics and Side-View 2D Video Analysis System Design

1. Kinetic Chain Architecture and Phase Segmentation in Infield Throwing

Baseball throwing mechanics operate through a proximal-to-distal kinetic chain, transferring mechanical energy generated by lower-extremity translation through the pelvis, thorax, shoulder girdle, and upper extremity to the baseball¹. While pitching leverages a downward mound slope to maximize forward linear displacement and peak ball velocity, an infielder's throw must be executed under distinct operational constraints: transitioning from fielding posture, executing rapid footwork, and stabilizing irregular centers of gravity³.

Biomechanical comparative studies indicate that compared to pitchers, infielders demonstrate a shorter stride phase duration, reduced pelvic backward rotation at foot strike, and an earlier temporal emergence of peak pelvic and trunk rotational velocities³. Achieving throwing speed and target accuracy within fractions of a second requires segmenting the motion into discrete temporal phases and defining spatial alignments at each phase boundary². From a biomechanical perspective, youth infield throwing is structured into six phases¹:

1. **Catch & Step Phase:** From the completion of fielding the ball to when the pivot foot (right foot for right-handed throwers) steps toward the target, placing the athlete's center of gravity firmly over the stance base of support⁶.
2. **Stride Phase:** The linear translation phase wherein the stride foot advances directly toward the target while the pivot hip maintains flexion and tension, concluding the exact frame the lead foot makes full ground contact (Foot Contact / Foot Strike)².
3. **Arm Cocking Phase:** From stride foot contact until the throwing shoulder achieves maximum external rotation (MER), driven by rotational separation between the pelvis and thorax¹.
4. **Arm Acceleration Phase:** From MER to the instant just prior to ball release, characterized by rapid forward trunk flexion, lead knee extension blocking, and explosive internal humeral rotation¹.
5. **Ball Release:** The temporal release instant where forward trunk tilt, elbow height, and elbow flexion angles dictate terminal ball speed, trajectory, and release consistency².
6. **Deceleration & Follow-Through Phase:** From release until the throwing shoulder completes internal rotation and the arm swings diagonally across the body, safely dispersing vast kinetic energy while resetting the fielder into an athletic posture².

Phase Name	Temporal Boundary Definition	Biomechanical Requirements	Key Kinematic Parameters
Catch & Step	Ball glove contact to pivot foot plant ⁶	Advance pivot foot under or ahead of fielding point; align head directly above pivot base of support ⁷ .	Pivot hip flexion angle, horizontal head displacement ¹
Stride	Weight gathering on pivot foot to lead Foot Contact (FC) ²	Linear translation toward target without premature pelvic opening; symmetrical arm separation ¹ .	Stride length (% of height), stride phase duration ¹
Arm Cocking	Foot Contact to Maximum External Rotation (MER) ¹	Develop torso-pelvis rotational separation; maintain elbow flexion at right angle in scapular plane ¹ .	Maximum shoulder ER, elbow flexion angle, hip-shoulder separation ¹
Arm Acceleration	MER to immediate pre-ball release ¹	Lead knee extension to brake linear momentum; explosive forward trunk tilt delivering the upper limb ¹ .	Peak forward trunk tilt velocity, lead knee extension angular velocity ¹
Ball Release	Instant of ball release ²	Upper arm aligned with bilateral shoulder axis (zero position); forward trunk tilt between 30 and 50 deg ¹¹ .	Forward trunk tilt, shoulder abduction angle, elbow height relative to torso ²

Follow-Through	Ball release to complete arm deceleration ⁵	Diagonal arm path across torso; deep trunk flexion and lead hip absorption to dissipate distraction forces ² .	Maximum trunk flexion angle, arc of arm deceleration, dynamic balance ²
-----------------------	--	---	--

2. Quantitative Kinematic Parameters for Lateral (Side-View) 2D Video Analysis

In single-camera 2D optical motion analysis, orienting the camera optical axis perpendicular (90 deg) to the infielder's throwing line provides an optimal sagittal plane perspective¹⁷. This perspective captures linear translation, joint flexion/extension, trunk tilt, and anterior-posterior displacement with minimal parallax and projective distortion¹⁷. Clinical assessment tools such as the Assessment of Biomechanical Efficiency System (ACES) rely on lateral views to screen kinetic chain efficiency and joint pathology risks².

Core anatomical landmarks tracked from the side view include the head (auditory meatus), torso (acromion process, greater trochanter), upper extremity (lateral humeral epicondyle, radial styloid process), and lower extremity (lateral knee joint line, lateral malleolus, fifth metatarsal head)¹⁸.

Stride Length Ratio

Calculated at the frame of stride foot contact as the horizontal pixel distance between the pivot foot toe and the stride foot heel, divided by the player's calibrated standing height pixel measurement². While pitchers aim for 75% to 85% of body height, infielders require a compact stride of 60% to 75% of body height to enable rapid release². A stride below 60% yields insufficient linear momentum, forcing compensatory arm throwing, whereas exceeding 75% physically impedes rapid pelvic rotation³.

Elbow Flexion Angle at Foot Contact

The sagittal plane interior angle formed between the upper arm vector (acromion to lateral epicondyle) and forearm vector (lateral epicondyle to radial styloid) at initial foot contact². The normative target is 70 deg to 95 deg². Arriving at foot contact with an extended elbow (> 95 deg) increases the moment of inertia of the arm, producing dangerous valgus torque on the ulnar collateral ligament (UCL) during subsequent external rotation².

Elbow Height at Ball Release

Assessed at release as the vertical orientation of the humerus relative to the torso axis line (line connecting the acromion and greater trochanter)². Mechanically, placing the humerus within the scapular plane with approximately 90 deg abduction constitutes the anatomical "zero position," balancing tension across the rotator cuff and minimizing capsular shear¹². In a 2D lateral frame, the elbow should align within -5 deg to +15 deg of a perpendicular line drawn from the trunk axis⁵.

Forward Trunk Tilt

Calculated as the angle formed between a vertical reference line and the torso vector (greater trochanter to acromion process)². The trunk should demonstrate a slight forward tilt of 5 deg to 15 deg at foot contact, progressing into a forward tilt of 30 deg to 50 deg at ball release².

Release trunk tilt directly correlates with terminal throw velocity and creates clearance for safe arm deceleration².

Lead Knee Extension Dynamics

The angular displacement of the stride knee from foot contact through release¹. After touching down at 130 deg to 140 deg of flexion, the lead knee must act as a firm mechanical brake, extending by +5 deg to +20 deg into release¹. This knee-brace mechanism arrests forward linear momentum and channels ground reaction forces into torso rotational acceleration¹.

Head Position Stability

The horizontal tracking of the ear landmark from fielding through release². During the step phase, the head must remain directly above the pivot foot base of support to establish potential energy¹. Premature forward drift of the head prior to foot contact is classified as "lunging," disrupting kinetic sequencing¹.

Kinematic Metric	Evaluation Phase	Operational Calculation	Normative Standard	Error Boundary
Stride Length Ratio	Lead Foot Contact ²	(Lead Heel X - Stance Toe X) / Height Pixels ²	60% to 75% of height ²	< 50% or > 85% of height ²
Elbow Flexion at FC	Lead Foot Contact ²	Angle: Acromion-Elbow-Wrist ²	70 deg to 95 deg ²	< 60 deg or > 115 deg ²
Elbow Height at Release	Instant of Ball Release ¹²	Humeral deviation from trunk perpendicular ¹ 2	-5 deg to +15 deg vs trunk normal ⁵	<= -16 deg (dropped elbow) ¹²
Forward Trunk Tilt	Instant of Ball Release ²	Trunk vector angle relative to vertical ²	30 deg to 50 deg ²	< 20 deg (upright) or > 60 deg ²
Lead Knee	Foot Contact to Release ¹	Release knee angle minus FC	+5 deg to +20 deg	<= -5 deg (knee

Delta		knee angle ¹	(extending) ¹	collapsing) ²
Head Drift	Mid-Stride Phase ¹	Ear X coordinate vs stance heel vertical line ¹	Aligned over pivot foot support ¹	Anterior drift past stance base ¹

3. Developmental Biomechanical Risks and Injury Prevention Criteria in Youth Players

During the adolescent growth spurt, rapid skeletal elongation outpaces muscular and tendinous adaptation, leaving open epiphyseal plates and articular cartilage vulnerable to repetitive shear stresses²³. Repeating mechanically inefficient throwing patterns during this growth window can cause career-threatening structural breakdown²³.

Medial baseball elbow (medial epicondyle apophysitis or avulsion fracture) stems from extreme valgus torque during late cocking and acceleration, pulling aggressively on the medial collateral ligament complex and immature growth plates¹⁴. Concurrently, compressive forces on the lateral radiohumeral joint produce osteochondritis dissecans (OCD) of the humeral capitellum, a severe cartilage lesion²³. At the shoulder, excessive rotational torque and distraction stresses can result in proximal humeral epiphysiolysis ("Little League shoulder")¹⁵.

In amateur coaching, "dropping the elbow" is widely recognized as a major technical fault¹².

However, the common verbal cue "raise your elbow high" can be counterproductive¹². In functional shoulder anatomy, aligning the humerus within the plane of the scapula at approximately 90 deg abduction represents the "zero position," where rotator cuff tension is evenly balanced and glenohumeral shear is minimized¹².

Instructing an athlete to lift the elbow above the shoulder line without sufficient trunk tilt compresses the supraspinatus tendon and subacromial bursa against the acromion, causing impingement and cuff tears¹². The visual appearance of a "low elbow" is typically not an isolated upper-limb failure, but a secondary breakdown of the kinetic chain—caused by insufficient stride length, premature pelvic rotation, or an upright torso²². Automated 2D analysis tools must evaluate elbow height as an angle relative to the torso axis, avoiding absolute height misinterpretations².

4. Common Infield Throwing Errors: Kinematic Signatures, Mechanisms, and Corrective Coaching Arm Throwing (Lack of Lower-Body Kinetic Chain Transfer)

Identified in lateral video by a narrow stride (< 50% body height) and immediate arm forward whip occurring concurrently with foot contact². Because the pivot hip fails to hinge and load, no hip-shoulder rotational separation is established².

The player cannot recruit force from the gluteal and core musculature, forcing the anterior

shoulder and elbow flexors to generate ball velocity in isolation². This spikes UCL valgus load and anterior capsular stress¹⁰.

- **Corrective Drill:** *Step-and-Hold Isometric Drill*⁴. From fielding posture, have the athlete plant the pivot foot, load the back hip, and hold balance for 1 second before stepping forward on a "one-two" rhythm⁴. Incorporating single-leg hops on the pivot foot helps develop lower-extremity drive²⁷.

Dropped Elbow (Deviation from Zero Position)

Identified at release when the elbow drops ≥ 16 deg below the line perpendicular to the torso axis¹². The forearm presents nearly parallel to the playing surface, resulting in a pushing arm action¹³.

This deviation places severe tensile strain on the medial UCL and epiphyseal plate¹⁴.

- **Corrective Cue & Drill:** Avoid telling players to "lift the elbow"²⁶. Provide an external focus cue: *"Release the ball from the height of your cap brim"*²⁶. This cue promotes forward trunk tilt while naturally aligning the arm into the zero position²². Implement the *Wall Shadow Throwing Drill*: stand with the throwing side along a wall; during the arm swing, keep the back of the hand and elbow gliding along the wall to maintain scapular plane alignment²⁸.

Premature Head Lunging (Forward Center of Gravity Collapse)

Identified during stride when the ear landmark shifts forward past the pivot foot base of support well before lead foot contact¹. Because youth athletes have a higher head-to-body mass ratio, forward head drift collapses back hip stability and dissipates ground reaction forces¹.

The torso lands excessively forward-flexed before rotation begins, locking thoracic mobility and forcing the arm to drag behind the torso¹.

- **Corrective Drill:** *Head-Pivot Vertical Alignment Drill*⁷. Instruct the fielder to place the pivot foot under the gloved ball upon fielding, keeping the head directly over the stance foot¹. The player lifts the lead knee and pauses briefly to confirm a balanced posture before striding linearly toward the target¹.

Premature Torso Opening (Early Chest Exposure)

Identified at initial foot contact when the chest and acromion rotate prematurely toward the side-view camera². The glove arm pulls open early, failing to maintain scapular retraction².

This error destroys hip-shoulder separation and leaves the throwing arm in hyperangulation (excessive horizontal abduction), placing dangerous tensile load on the anterior glenohumeral capsule².

- **Corrective Drill:** *Glove-Shoulder Swap Drill*³¹. Have the player keep the glove shoulder pointed directly at the target through foot contact, rotating the glove shoulder backward at landing to whip the throwing shoulder forward³¹. Using a slightly closed stride foot placement helps prevent premature pelvic opening².

Lead Knee Collapse (Loss of Braking Support)

Identified from foot contact through release when the lead knee continues to flex and drift forward, failing to achieve extension¹.

Without lead knee extension, linear momentum is absorbed in the quad rather than converted into trunk rotational acceleration¹. Release velocity drops, and ball trajectory flattens prematurely¹¹.

- **Corrective Drill: Front Knee Stopper & Freeze Drill²**. Place a low barrier in front of the lead foot to prevent the knee from drifting forward past the toes. Have the athlete finish throws by "freezing" balanced entirely on the lead leg with the knee firm, reinforcing eccentric quad and gluteal braking².

Error Pattern	Lateral 2D Visual Signature	Biomechanical Mechanism	Practical Drill & Verbal Cue
Arm Throwing	Stride < 50% height; arm whips forward immediately at contact ² .	No pelvic-torso separation; arm operates isolated from lower body, increasing UCL torque ² .	Step-and-Hold Drill Load pivot hip for 1 second, then drive forward on a "one-two" count ⁴ .
Dropped Elbow	Elbow drops ≥ 16 deg below torso perpendicular at release ¹² .	Zero-position breakdown; humerus pushes forward, generating excessive valgus stress ¹² .	Wall Shadow Throwing Cue: "Throw from cap brim height"; trace arm along a wall ²⁶ .
Head Lunging	Head landmark drifts past stance foot before foot contact ¹ .	Forward head drop causes early pelvic opening, locking thoracic rotation ¹ .	Head-Pivot Alignment Plant pivot foot under fielding glove; balance head over stance foot before stride ¹ .
Early Opening	Chest faces camera at foot contact; glove arm drops	Loss of rotational elastic energy; shoulder dragged	Shoulder Swap Drill

	early ² .	into hyperangulation ² .	Keep glove shoulder on target until contact, then swap shoulder positions ³¹ .
Knee Collapse	Lead knee angle decreases (flexes) from contact to release ¹ .	Ground reaction braking lost; linear energy absorbed instead of rotating trunk ¹ .	Front Knee Stopper & Freeze Block knee forward drift; hold a stable one-leg finish after release ² .

5. 2D Lateral Video Analysis Tool: Scoring Algorithm and Feedback Generation Logic

The automated video evaluation tool processes 2D coordinate time-series generated by pose estimation frameworks (such as MediaPipe) using a deduction-based scoring system².

5.1 100-Point Scoring Model

The overall score begins at 100 points. Penalties are deducted based on kinematic deviations across six core lateral indicators, weighted (w_i) according to injury prevention priority and mechanical efficiency²:

$$Score = 100 - \sum_{i=1}^6 w_i \cdot P_i$$

The penalty factor P_i is assigned as 0 for within-norm performance, 0.5 for moderate error, and 1.0 for severe mechanical deviation.

Weight distribution allocates 25 points to elbow height at release due to its direct association with UCL valgus loading and Little League elbow¹⁴. The remaining five parameters (head stability, stride ratio, elbow flexion at contact, trunk tilt, and knee extension) are allocated 15 points each¹.

Item ID	Kinematic Indicator	Phase Evaluated	Normal Range (P = 0)	Minor Error (P = 0.5)	Major Error (P = 1.0)	Weight
---------	---------------------	-----------------	----------------------	-----------------------	-----------------------	--------

Item 1	Head Stability	Mid-Stride Phase	Head X within +/-10% of pivot heel vertical ¹	Premature drift (11% to 25% deviation) ¹	Severe lunge (> 25% forward drift) ¹	15 pts
Item 2	Stride Length Ratio	Foot Contact	60% to 75% standing height ²	50% to 59% or 76% to 85% of height ²	< 50% or > 85% of height ²	15 pts
Item 3	Elbow Flexion at FC	Foot Contact	70 deg to 95 deg flexion ²	96 deg to 115 deg flexion ²	> 115 deg or < 60 deg ²	15 pts
Item 4	Elbow Height at Release	Ball Release	-5 deg to +15 deg vs torso normal ⁵	-6 deg to -15 deg vs torso normal ¹³	<= -16 deg (dropped elbow) ¹²	25 pts
Item 5	Forward Trunk Tilt	Ball Release	30 deg to 50 deg forward ²	20 deg to 29 deg or 51 deg to 60 deg ²	< 20 deg (upright) or > 60 deg ²	15 pts
Item 6	Lead Knee Extension	Foot Contact to Release	Knee angle delta >= +5 deg ¹	Angle delta between -5 deg and +5 deg ²	Knee angle delta <= -5 deg (collapsing) ²	15 pts

5.2 Automated Feedback Generation Logic

When the system registers a penalty ($P_i \geq 0.5$), it generates a targeted feedback message combining three elements:

1. **Quantitative Assessment:** The measured metric and its deviation from standard.
2. **Biomechanical Rationale:** Why the deviation impairs performance or elevates injury risk.
3. **Actionable Correction:** An intuitive movement cue and a targeted drill.

Example Output (Item 4 - Dropped Elbow Triggered):

- **Measured Value:** Elbow position is -18 deg relative to trunk perpendicular (Dropped

Elbow detected).

- **Why It Matters:** Releasing with the elbow below the shoulder-torso axis pulls the arm out of the zero position, sharply increasing valgus torque on the medial elbow ligament and growth plate¹².
- **How to Fix:** Focus on tilting the upper body forward while throwing from "the height of your cap brim"²². Practice the *Wall Shadow Throwing Drill* (10 reps x 3 sets) to groove proper scapular plane arm elevation²⁸.

6. Optical Setup, Camera Calibration, and Implementation Requirements

The accuracy of 2D kinematic measurements depends on camera positioning and optical geometry¹⁷. To minimize measurement error during field capture, the following recording protocol should be maintained:

1. **Perpendicular Optical Alignment:** The camera must be placed at a strict 90 deg angle relative to the thrower's directional vector (the line between fielding point and target)¹⁸. Angular misalignment introduces parallax error, distorting sagittal joint angles and stride length calculations¹⁸.
2. **Mounting Height and Elevation:** Position the lens between 0.9 m and 1.1 m above ground level (chest-to-hip height of youth players) with a level optical axis. Tilting the camera up or down distorts vertical coordinates and trunk tilt angles.
3. **Stand-Off Distance:** Place the camera 4.5 m to 6.0 m away from the player. This distance accommodates the entire movement arc while reducing perspective distortion between near and far limbs¹⁸.
4. **Frame Rate and Shutter Speed:** Record at a minimum of 120 fps, with 240 fps recommended¹⁸. Internal shoulder rotation velocities exceed 7,000 deg/second during release, making standard 30 fps video susceptible to severe motion blur and missed release frames¹⁴. Configure the shutter speed between 1/500 and 1/1000 second¹⁸.
5. **Spatial Calibration:** Calibrate pixel coordinates using a 1.0 m vertical reference object placed in the action plane or normalize distances against the subject's standing height pixels².

Parameter	Recommended Specification	Optical Quality Rationale
Camera Angle	Strictly 90 deg perpendicular to throw vector ¹⁸	Eliminates parallax distortion of sagittal joint angles ¹⁸
Camera Height	0.9 m to 1.1 m (chest level, level horizon)	Prevents projection skew from elevated or low

		camera angles
Distance from Subject	4.5 m to 6.0 m	Minimizes perspective distortion while keeping all phases in frame ¹⁸
Frame Rate	120 fps to 240 fps (minimum 60 fps) ¹⁸	Resolves rapid arm acceleration (> 7,000 deg/s) without dropped keyframes ¹⁴
Shutter Speed	1/500 s to 1/1000 s [cite: 18]	Prevents motion blur at the wrist, hand, and baseball ¹⁸
Spatial Scaling	1.0 m scale object or standing height normalization ²	Converts pixel distances into normalized biomechanical units ²

7. Conclusion

Lateral (side-view) 2D video analysis provides a practical, accurate method for evaluating kinetic chain efficiency and joint loading in youth infielders¹⁷. Unlike pitching from a mound, infield throwing requires rapid force generation through a compact stride following the catch³. The standard benchmarks for youth infield throwing include:

- A stable head position over the pivot foot during the step¹;
- A compact stride length of 60% to 75% body height²;
- An elbow flexion angle between 70 deg and 95 deg at foot contact²;
- Humeral alignment in the anatomical zero position at release¹²;
- Forward trunk tilt of 30 deg to 50 deg at release²; and
- Firm lead knee extension to convert forward momentum into rotational power¹.

Combining a 100-point deduction model with targeted, biomechanically sound feedback provides an objective foundation for software development, helping coaches avoid risky cues and supporting safe athletic development¹².

引用文献

1. Biomechanical Analysis of the Throwing Athlete and Its Impact on, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8811517/>
2. <https://ijspt.scholasticahq.com/article/29869-reliability-of-an-observational-biomechanical-analysis-tool-in-adolescent-baseball-pitchers/attachment/75413.pdf>
3. Kinematic comparison of baseball delivery with regard to trunk and, https://www.researchgate.net/publication/282486415_Kinematic_comparison_of_baseball_delivery_with_regard_to_trunk_and_upper_limb_motions_for_the_infield

[er among various levels of development](#)

4. 一塁送球ミスを減らす実践ドリル6選 - 野球 - AIスポーツトレーナー,
<https://ai-sportstrainer.com/columns/baseball-infield-throwing-stability-step>
5. Baseball | Musculoskeletal Key, <https://musculoskeletalkey.com/baseball-2/>
6. ソフトボール外野手のバックホーム返球時間 に及ぼす下肢ステップ,
<https://www3.sonoda-u.ac.jp/tosyo/ronbunsyu/%E5%9C%92%E7%94%B0%E5%A%D%A6%E5%9C%92%E5%A5%B3%E5%AD%90%E5%A4%A7%E5%AD%A6%E8%A%B%96%E6%96%87%E9%9B%8648/043-052.PDF>
7. 守備職人が教える守備ドリル 送球を安定させるポイントは頭の位置,
<https://fmv-mypage.fmworld.net/fmv-sports/post-16300/>
8. 【内野ゴロの送球ステップ】お父さんとお母さんのための野球基礎,
<https://baseballking.jp/yakyuiku/181794/>
9. 【スポーツ応用編】野球肘を予防するためのピッチングフォームの,
<https://arch-seminar.com/posturalanalysis/%E3%80%90%E3%82%B9%E3%83%9D%E3%83%BC%E3%83%84%E5%BF%9C%E7%94%A8%E7%B7%A8%E3%80%91%E9%87%8E%E7%90%83%E8%82%98%E3%82%92%E4%BA%88%E9%98%B2%E3%81%99%E3%82%8B%E3%81%9F%E3%82%81%E3%81%AE%E3%83%94%E3%83%83/>
10. 投球フォームの徹底解析 | 広島のスポートス整体フィジカルデザイン,
<https://physical-design.net/free/throw-form-and-mechanism>
11. KINEMATIC ATTRIBUTES ASSOCIATED WITH OVERARM,
<https://commons.nmu.edu/cgi/viewcontent.cgi?article=2241&context=isbs>
12. 少年野球で球速アップ！元補欠が明かす「自宅で我が子をエースに」,
<https://www.elevendesign.online/archives/20136>
13. 内野手 送球においてのポイント - トップの作り方編 - |,
<https://baseball-one.com/blog/archives/183525/>
14. Biomechanics of Pitching: Pushing Limits on the Shoulder and Elbow,
<https://sites.nd.edu/biomechanics-in-the-wild/2019/09/16/biomechanics-of-pitching-pushing-limits-on-the-shoulder-and-elbow/>
15. Adolescent Baseball Pitching Technique: A Detailed Three,
https://www.researchgate.net/publication/6078670_Adolescent_Baseball_Pitching_Technique_A_Detailed_Three-Dimensional_Biomechanical_Analysis
16. 投球障害・野球肘の見つけ方 - BIGLOBE,
<https://www5d.biglobe.ne.jp/wakata/daihyo/zasshi/journal0612/toukyuusyougai2.html>
17. Validity and Reliability of a Two-Dimensional Video Analysis Method,
<https://www.mdpi.com/2076-3417/9/19/3975>
18. Kinematic Movement Analysis: Part One - Sports Injury Bulletin,
<https://www.sportsinjurybulletin.com/improve/tools-and-technology/kinematic-movement-analysis-part-one>
19. Reliability of an Observational Biomechanical Analysis Tool in,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8637241/>
20. measurement of shoulder range of motion, video analysis of pitching,
<https://twu-ir.tdl.org/bitstreams/c0f88821-e61f-44ec-8f67-f89df0d76288/download>

21. 理想の投球フォームの科学～球速と障害予防の両立,
<https://www.sept-kiraku.com/%E3%82%B3%E3%83%A9%E3%83%A0-1/%E7%90%86%E6%83%B3%E3%81%AE%E6%8A%95%E7%90%83%E3%83%95%E3%82%A9%E3%83%BC%E3%83%A0/>
22. 「肘を上げろ」で野球肘が悪化？ 中学2年生の事例から学ぶ,
<https://www.elevendesign.online/archives/19430>
23. 栃木県青少年野球団体協議会 野球手帳,
https://www.tochigi-koyaren.net/info/pdf/yakyu_techou_2025.pdf?v=1024
24. Reliability of an Observational Biomechanical Analysis Tool in,
<https://ijspt.scholasticahq.com/article/29869-reliability-of-an-observational-biomechanical-analysis-tool-in-adolescent-baseball-pitchers>
25. ジュニア(小中学生)層に関する基本方針,
https://www.shibata-sports.jp/editor/upload/files/O2_3.%E3%82%B8%E3%83%A5%E3%83%8B%E3%82%A2%E5%B1%A4%E6%96%B9%E9%87%9D%E7%BC%88%E6%96%B0%E6%BD%9F%E7%9C%8C%E8%BB%9F%E5%BC%8F%E9%87%8E%E7%90%83%E9%80%A3%E7%9B%9F%E3%82%88%E3%82%8A%E7%BC%89%28%29.pdf
26. 【少年野球】お母さんでも分かる「投球時のヒジの高さ」,
<https://www.89pspot.com/entry/2024/07/02/201641>
27. 守備練習 間違いがちなバックホームへの送球ステップ 足の角度,
<https://fmv-mypage.fmworld.net/fmv-sports/post-15631/>
28. 野球のきれいな投げ方を身につける | 肩・肘に負担の少ない,
<https://ai-sportstrainer.com/columns/clean-throwing-form>
29. 【YAKYUNOTE編集長が語る】少年野球指導の「心・技・体」,
<https://baseballnote.com/junior-baseball-coaching-guide/>
30. How to Create High-Velocity Softball Throwing Mechanics in 2022,
<https://danblewett.com/guide-fastpitch-softball-throwing-mechanics/>
31. 内野手 送球におけるポイント -ステップから送球編- |,
<https://baseball-one.com/blog/archives/183604/>
32. 「腕を振れ」では伝わらない？ 重要なグラブ側の肩...ボールの強さ,
<https://full-count.jp/2025/12/02/post1872267/>
33. Phases of The Overhand Throw Reprinted from Journal of,
https://www.researchgate.net/figure/Phases-of-The-Overhand-Throw-Reprinted-from-Journal-of-Biomechanics-Vol-32-Fleisig-GS_fig1_356723541